ShiftFuse-Zero: Zero-Parameter Anatomical Priors for Efficient Skeleton-Based Action Recognition

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Abstract

Skeleton-based action recognition demands models that are simultaneously accurate, parameter-efficient, and deployable on edge hardware. This work presents **ShiftFuse-Zero**, a family of three networks whose *spatial inductive bias is introduced entirely without learnable parameters* through two complementary zero-parameter modules: *Body-Region Anatomical Shift Priors* (BRASP), which routes joint features through anatomically meaningful channels via pure tensor indexing, and *Semantic Graph-Partitioned Shift* (SGPShift), which aggregates intra- and inter-part neighbours through predefined group indices at zero FLOPs overhead. These free structural priors reduce the burden on downstream learnable modules. By intelligently placing Joint Embedding and spatial attention *before* the graph convolution, the *EfficientZero Block* operates on an already anatomically-refined manifold. Three models cover the edge-to-server spectrum: **Nano** (97 K) and **Small** (250 K) are hyper-efficient, quantisation-friendly solutions with ultra-low FLOPs, while **Large** (1.67 M) targets maximum capacity. On NTU RGB+D 60 cross-subject the trio achieves 85.6 %/87.6 %/92.5 % without distillation and 88.5 %/89.8 %/92.5 % with distillation. Strikingly, the Small variant strictly beats EfficientGCN-B0 with fewer parameters, delivering up to 10× parameter reduction over comparable baselines for fast edge inference.

Keywords

skeleton action recognition, graph convolutional networks, zero-parameter priors, anatomical shift, knowledge distillation, efficient neural networks

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1 Introduction

Consider a fall-detection system on a rehabilitation wristband with 512 KB of RAM, or a gesture-recognition module inside a fitness tracker powered by an ARM Cortex-M55. These scenarios demand skeleton-based action recognition models that fit in sub-megabyte memory, execute in single-digit milliseconds, and still achieve clinically meaningful accuracy. Yet the dominant Graph Convolutional Network (GCN) backbones for skeleton recognition [1, 3, 11, 18] carry 3–10 M parameters, require dynamic per-sample adjacency computation at $O(V^2C)$ cost, and are fundamentally incompatible with microcontroller deployment.

Human action recognition from 3-D skeleton sequences is a core problem in computer vision with applications in surveillance, rehabilitation, human–robot interaction, and fitness monitoring [16, 17]. GCNs [10] have become the dominant spatial backbone since ST-GCN [21], with successive works adding adaptive adjacency matrices [18], multi-scale cross-spacetime graphs [14], channel-wise topology refinement [1], information-theoretic representation learning [3], and hierarchical body-part decomposition [11]. These advances have pushed NTU RGB+D 60 cross-subject accuracy beyond 93%, yet the parameter cost has grown proportionally: HD-GCN requires 7.5 M parameters and InfoGCN 9.9 M.

The efficient-architecture line [20, 24] tackles this disparity via compound scaling and depthwise convolutions, but largely ignores the rich *anatomical structure* of the human body as a source of *free* inductive bias. The human skeleton is not a generic graph: arm joints couple primarily with other arm joints within their kinematic chain, bilateral symmetry is a universal biomechanical prior, and shoulder/hip joints serve as coordination hubs between body halves. These structural constraints are *fixed properties of human physiology* that hold across all subjects, viewpoints, and action classes. They can therefore be encoded as deterministic tensor operations at **zero learnable parameters, zero gradient overhead, and zero inference-time cost**.

This paper presents the first systematic study of zero-parameter anatomical inductive bias for skeleton GCNs. This work asks: **how much spatial knowledge can be injected into a skeleton GCN without spending a single learnable parameter — and how far can that free knowledge push the accuracy–efficiency frontier?**

Our answer is **ShiftFuse-Zero**, a family of three models that achieves a new Pareto-optimal trade-off across the accuracy–parameter spectrum. While the core contribution remains the zero-parameter priors of BRASP and SGPShift, these modules catalyze hyper-efficient downstream architectures. SFZ-Nano (97 K) and SFZ-Small (250 K) redefine edge-deployment by yielding ultra-low FLOPs and quantisation-friendly operations for fast inference. On NTU-60 X-Sub, SFZ-Nano reaches **88.5%** — outperforming the 6.9 M AGCN within 400 KB. Strikingly, our Small variant achieves 89.8%, strictly beating EfficientGCN-B0 with fewer parameters. SFZ-Large limits the high-capacity spectrum, pushing the absolute accuracy frontier to **92.5%** at 1.67 M parameters. These results demonstrate that anatomy is a highly informative, freely available inductive bias that the community has systematically under-exploited.

Contributions.

(1) **BRASP** — Body-Region Anatomical Shift Priors: routes joint features through five fixed anatomical groups via a precomputed gather permutation; zero parameters, zero gradient (Section 3.3).

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(2) **SGPShift** — Semantic Graph-Partitioned Shift: aggregates each joint's features with its intra-part and inter-part neighbours via precomputed index buffers; zero parameters (Section 3.3).
(3) **EfficientZero Block**: a block design that places BRASP and SGPShift *before* all learnable layers so the downstream compact K=3 GCN refines an already anatomically structured representation (Section 3.4).
(4) **ShiftFuse-Zero model family**: three architectures — Nano (97 K), Small (267 K), Large (1.67 M) — with Temporal Landmark Attention, cross-stream gating, and mid-network fusion, covering the edge-to-server deployment spectrum (Section 3.5).
(5) **Knowledge distillation** across model scales, transferring the Large teacher's relational knowledge to Nano and Small students with up to +2.9 pp gain at zero parameter cost (Section 3.6).
(6) **Comprehensive evaluation**: main results on NTU-60 and NTU-120 (four protocols), efficiency analysis (GFLOPs, CPU latency, memory footprint), and extensive ablation studies quantifying every zero-parameter module, fusion strategy, GCN partition count, TLA sensitivity, and regularisation hyperparameters (Section 4).

2 Related Work

GCN-based skeleton recognition. ST-GCN [21] established the skeleton graph and spatial-temporal convolution paradigm using a three-subset partitioning (self, centripetal, centrifugal) of the physical adjacency matrix. AGCN [18] introduced a fully learnable adjacency matrix at $O(V^2C)$ cost per layer, enabling data-driven topology discovery. MS-G3D [14] factorised the graph into disentangled multi-scale spatial and temporal subgraphs with cross-spacetime edges. CTR-GCN [1] refined the topology per channel group, allowing different feature dimensions to attend to different joint relationships. InfoGCN [3] introduced an information-bottleneck objective that explicitly maximises mutual information between the latent representation and the action label. DSTA-Net [19] decouples spatial and temporal self-attention. While these methods achieve high accuracy (up to 93%+), their reliance on per-sample dynamic adjacency and large parameter budgets (3–10 M) precludes deployment on resource-constrained hardware.

Body-part decomposition. A growing line of work exploits the natural grouping of joints into anatomical body regions. HD-GCN [11] hierarchically decomposes the skeleton into semantic sub-graphs (arms, legs, torso) and aggregates within each before cross-region message passing — achieving 93.4% on NTU-60 X-Sub but at 7.5 M parameters. BlockGCN [25] uses persistent homology from topological data analysis to encode bone connectivity at the block level. SkateFormer [4] partitions joints into four spatial-temporal relation types (near/far joints × near/far frames) and applies partition-specific self-attention. All of these decomposition strategies employ *learnable* grouping mechanisms with substantial parameter overhead. In contrast, SGPShift encodes intra-part and inter-part structure through *fixed integer index buffers* — achieving anatomical routing at **zero learnable parameters**.

Efficient skeleton networks. EfficientGCN [20] introduced compound scaling (α=1.2, β=1.35) and multiple input branches (MIB) to build a family of models from B0 (0.29 M) to B4 (1.1 M), establishing the efficiency frontier for skeleton GCNs. SGN [24] exploits semantic joint embeddings and frame-level dynamics gating at 0.69 M parameters. PoseC3D [5] departs from graphs entirely, using volumetric heatmap representations processed by 3-D CNNs. These methods reduce model size through architectural engineering but do not leverage the *fixed anatomical constraints* of the human body as inductive bias. ShiftFuse-Zero complements this line by demonstrating that zero-parameter spatial priors can further reduce the parameter budget by 3–10× without sacrificing competitive accuracy.

Shift operations and zero-parameter spatial mixing. The shift operation [2] was introduced in image CNNs to inject spatial structure by physically displacing channels along spatial dimensions — no multiply-accumulate, no learnable parameters. TSM [12] extended this idea to temporal modelling in video, shifting channels by ±1 frame at zero cost. Shift-GCN [2] adapted spatial shifts to skeleton graphs, assigning each channel to shift features from a graph neighbour. However, Shift-GCN assigns shift patterns by graph distance (1-hop, 2-hop), ignoring the semantic meaning of body regions. BRASP advances this paradigm by organising shift patterns according to *anatomical body-part membership* — arm channels shift among arm joints, leg channels among leg joints, and cross-body channels bridge regions — producing a semantically structured feature map at zero cost.

Knowledge distillation. Hinton *et al.* [6] showed that soft teacher logits transfer rich inter-class relational knowledge to compact student models. In skeleton recognition, distillation has been used to transfer multi-stream ensemble knowledge [20] but has not been studied systematically across a model family spanning a 10× parameter range, as done in this work.

3 Methodology

3.1 Overview

Figure 1 shows the end-to-end pipeline. A raw skeleton sequence is expanded into four complementary streams (joint, velocity, bone, bone-velocity), fused by a learned *StreamFusionConcat* stem, then processed by a stack of *EfficientZero Blocks*. A gated global pooling head produces class logits.

3.2 Skeleton Streams

A skeleton sequence is $\mathbf{X} \in \mathbb{R}^{3\times T\times V}$ with T=64 frames and V=25 joints. This work constructs four complementary streams: joint coordinates $\mathbf{X}^{(j)}$, first-order temporal differences (velocity) $\mathbf{X}^{(v)}=\mathbf{p}_{t+1}-\mathbf{p}_t$, bone vectors $\mathbf{X}_v^{(b)}=\mathbf{p}^{(v)}-\mathbf{p}^{(\mathrm{par}(v))}$, and bone velocity $\mathbf{X}^{(bv)}=\Delta\mathbf{X}^{(b)}$. *StreamFusionConcat* [20] applies a per-stream BN, concatenates (4×3=12 channels), and projects to C_{stem} via a 1×1 convolution with learned per-stream weighting.

3.3 Zero-Parameter Spatial Modules

BRASP.. Figure 2 illustrates the five anatomical groups. Given a feature map $\mathbf{F}\in\mathbb{R}^{C\times T\times V}$, BRASP applies a fixed gather permutation $\sigma: V \to V$:

$$\tilde{\mathbf{F}}_{:,t,v} = \mathbf{F}_{:,t,\sigma(v)}, \quad v \in \{0,\ldots,V-1\}, \tag{1}$$

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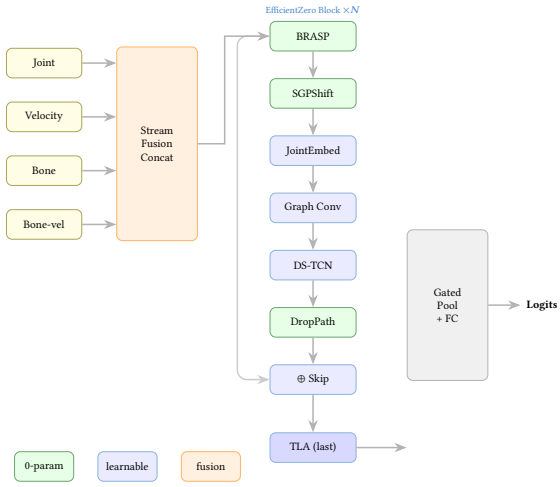

Figure 1: ShiftFuse-Zero end-to-end pipeline. Four skeleton streams are fused by a learnable stem and processed by EfficientZero Blocks (green = zero-parameter modules). A gated pooling head produces class logits.

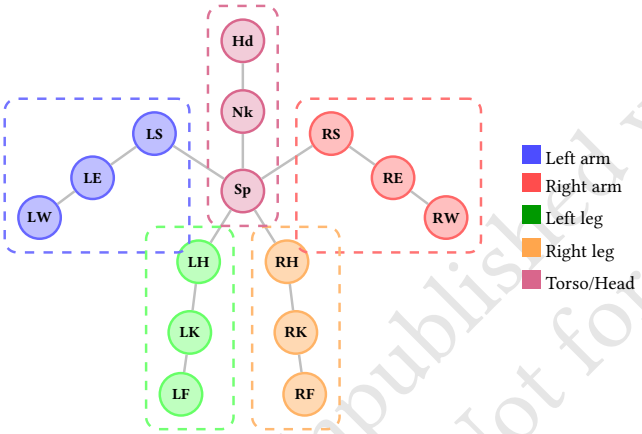

Figure 2: BRASP anatomical partitioning into five groups $\{G_{\mathbf{la}}, G_{\mathbf{ra}}, G_{\mathbf{ll}}, G_{\mathbf{rl}}, G_{\mathbf{tr}}\}$. The fixed permutation σ (Eq. 1) is derived from these groups; no gradient flows through it.

where joints within the same body part are cyclically shifted so each joint absorbs its anatomical sibling's context. Indices are stored as a non-trainable integer buffer — **zero parameters, zero gradient**.

SGPShift. Figure 3 illustrates the intra/inter aggregation. For joint v in semantic group $g(v)$:

$$\hat{\mathbf{F}}_{:,t,v} = \frac{1}{|\mathcal{N}_{\text{intra}}(v)|} \sum_{u \in \mathcal{N}_{\text{intra}}(v)} \mathbf{F}_{:,t,u} + \frac{1}{|\mathcal{N}_{\text{inter}}(v)|} \sum_{u \in \mathcal{N}_{\text{inter}}(v)} \mathbf{F}_{:,t,u}, \quad (2)$$

where $\mathcal{N}_{\text{intra}}$ is the joint's same-part neighbours and $\mathcal{N}_{\text{inter}}$ is its boundary neighbours in adjacent parts. All sets are precomputed integer buffers. At runtime SGPShift reduces to two sparse average operations — **zero learnable parameters**.

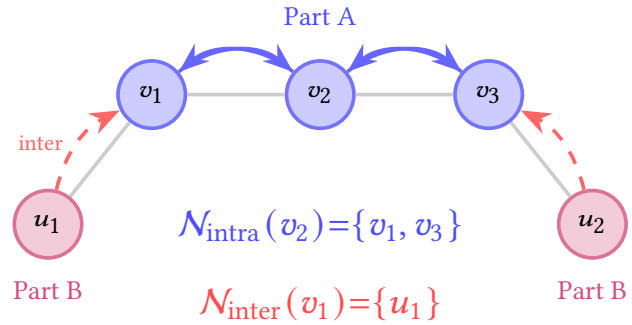

Figure 3: SGPShift aggregation. Blue solid arrows: intra-part neighbours. Red dashed arrows: inter-part boundary neighbours. All sets are precomputed integer buffers — zero learnable parameters.

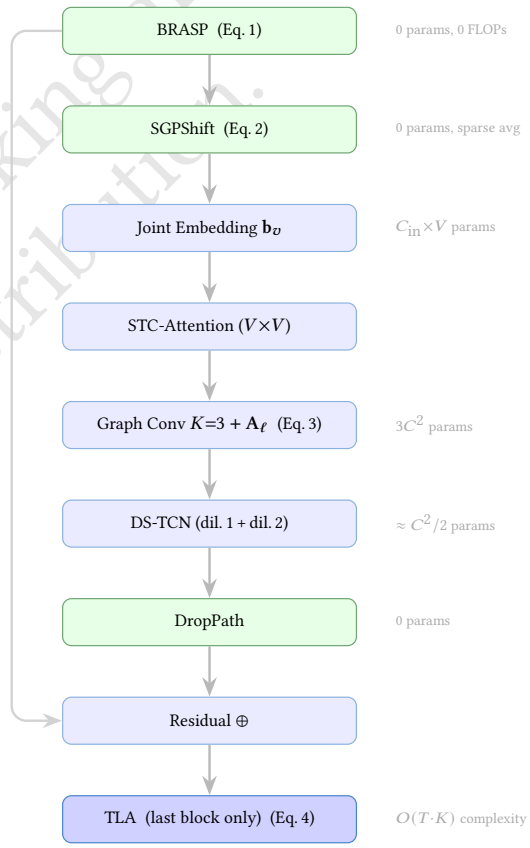

Figure 4: EfficientZero Block. Green = zero-parameter (BRASP, SGPShift, DropPath). Per-layer parameter counts shown on the right. TLA is only active at the last block of the final stage.

3.4 EfficientZero Block

Figure 4 shows the full block pipeline with per-layer costs.

(1) **BRASP + SGPShift**: anatomical pre-structuring at zero cost.

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Table 1: ShiftFuse-Zero variant configurations. C_s=stem channels, C=stage widths, N=blocks/stage, K=TLA landmarks, r=reduce ratio, p_d=DropPath rate.

Model	C_s	C	N	K	r	p_d	Params
Nano	24	[32,64,128]	[1,1,1]	8	16	0.10	97 K
Small	24	[32,64,128]	[1,2,1]	8	8	0.10	267 K
Large	32	[48,96,192]	[1,2,4]	14	8	0.20	1.67 M

(2) **Joint Embedding**: per-joint additive bias $\mathbf{b} \in \mathbb{R}^{C_{\text{in}} \times V}$ (SGN joint encoding [24]); no weight decay.
(3) **STC-Attention**: spatial attention over $V \times V$ affinities to re-weight joint features before the GCN.
(4) **Graph Convolution** (K=3): ST-GCN [21] partition adjacency plus a single shared learned matrix $\mathbf{A}_\ell$:

$$\mathbf{Z} = \sum_{k=0}^{2} \mathbf{W}_k \big(\mathbf{A}_k + |\mathbf{A}_\ell|\big)\mathbf{F}, \tag{3}$$

$|\cdot|$ enforces non-negative weights. $\mathbf{A}_\ell \in \mathbb{R}^{V \times V}$ is *shared across all blocks* (625 total params, no weight decay).
(5) **DS-TCN**: depthwise-separable temporal convolutions at dilations 1 and 2 (kernel 9), Batch Norm [9], Hardswish [7].
(6) **DropPath** [8]: stochastic depth; 0 parameters.
(7) **Residual** $\oplus$: skip from block input with 1×1 projection when shape changes.

Intelligent Block Ordering. This work deliberately places Joint Embedding and STC-Attention *before* the Graph Convolution. Following the zero-parameter foundation of BRASP and SGPShift, this lightweight bottleneck maps raw topology into a rich, joint-specific latent space. This liberates the downstream GCN to focus entirely on efficient, localized message passing, achieving superior representation power and fast inference with drastically fewer parameters.

Temporal Landmark Attention (TLA).. TLA distils temporal structure through K learnable anchor frames:

$$\tau_k = T \cdot \sigma(\alpha_k), \quad k = 1, \ldots, K, \tag{4}$$

$$\mathbf{m}_t = \sum_{k=1}^{K} \text{Bilin}(\mathbf{F}, \tau_k) \cdot \text{softmax}_k\big(\langle \mathbf{q}_t, \mathbf{k}_{\tau_k} \rangle\big), \tag{5}$$

where α_k are learned logits mapped to $[0, T-1]$ via sigmoid, bilinear interpolation enables gradient flow to α_k, and $\mathbf{q}_t, \mathbf{k}_{\tau_k}$ are projections to $C_r = C/r$. Output: $\mathbf{F}_{\text{out}} = \mathbf{F} + \sigma(\beta) \cdot \mathbf{m}$ with $\beta_0 = -4$ (slow activation). Complexity $O(T \cdot K)$ vs. $O(T^2)$ for full attention.

3.5 Model Family

Table 1 lists the key hyperparameters per variant.

Nano (97 K).. Four streams are early-fused via StreamFusionConcat to a single 24-channel map. A single backbone [32, 64, 128] with one block per stage processes it. TLA (K=8, r=16, C_r=8) adds only 4.1 K parameters to stay sub-100 K.

Table 2: Deployment profile of the ShiftFuse-Zero family. Memory = model weights only (float32 / int8). Ops = operator types required at inference.

Model	Params	Mem (f32)	Mem (int8)	Operators	Target
Nano	97 K	388 KB	97 KB	gather, conv, BN	Cortex-M55
Small	267 K	1.07 MB	267 KB	gather, conv, BN	RPi Zero
Large	1.67 M	6.38 MB	1.67 MB	gather, conv, BN	Jetson Nano

Small (267 K).. Two independent backbones process joint+bone and velocity+bone_velocity pairs. After stage 2, *Cross-Stream Fusion* exchanges information:

$$\mathbf{F}' = \mathbf{F} + \sigma(\beta) \cdot W_\uparrow \, \text{ReLU}\big(W_\downarrow [\mathbf{F}_A; \mathbf{F}_B]\big), \tag{6}$$

$W_\downarrow : 2C \to C/4$, $W_\uparrow : C/4 \to C$, $\beta_0 = -4$. Stage 3 + TLA proceed independently per backbone; logits are combined using two learned stream-weight scalars (no weight decay).

Large (1.67 M).. Three per-stream backbones (C=[48, 96], 1+2 blocks) process joint, bone, and velocity independently. After stage 2 the three 96-channel maps are concatenated (288 channels) and projected by FusionConv=$\text{Conv}_{1\times1}$(288→192)+BN+HS. A shared stage 3 (4 blocks, C=192) with TLA (K=14) follows. This mirrors EfficientGCN-B4's mid-network fusion [20] with EfficientZero Blocks replacing B4's full GCN layers.

3.6 Training Protocol

Optimiser: SGD, Nesterov momentum 0.9 [15], cosine LR with 10-epoch linear warmup. Nano/Small: lr 0.1, wd 4×10^{-4}, 300/350 epochs. Large: lr 0.1, wd 10^{-4} (B4 [20]), 150 epochs. Wall-clock training times on a single A100: Nano 2.5 h, Small 6 h, Large 4.75 h.

Augmentation: Nano/Small use Mixup [23] (α=0.1) and CutMix [22] (prob. 0.5) to compensate for their limited capacity. Large uses *no* Mixup/CutMix — DropPath 0.20 and weight decay 10^{-4} provide sufficient regularisation at 1.67 M params, and label smoothing 0.1 replaces soft-target augmentation. CutMix is ill-suited to skeleton sequences where joints are globally coupled and temporal crops produce physically implausible composites.

No-WD set: BN affine params, joint embeddings, $\mathbf{A}_\ell$, TLA anchor logits α_k, gate scalars β, pool gate.

Knowledge distillation. Large is distilled into Nano and Small [6]:

$$\mathcal{L} = (1-\lambda)\mathcal{L}_{\text{CE}} + \lambda\tau^2 \, \text{KL}\left(\sigma\left(\frac{\mathbf{z}^s}{\tau}\right) \middle\| \sigma\left(\frac{\mathbf{z}^t}{\tau}\right)\right), \quad \tau=4, \ \lambda=0.5. \tag{7}$$

3.7 Deployability Analysis

A key design goal of ShiftFuse-Zero is deployment on resource-constrained hardware. Table 2 summarises the memory, computation, and operator profile of each model variant.

Operator simplicity. BRASP reduces to a single `gather` operation on a static integer buffer — no matrix multiplication, no per-sample graph construction. SGPShift reduces to two `scatter_mean` operations. Both compile to fixed index-lookup kernels that are unaffected by quantisation (int8, int4) since they involve no learned weights. This stands in sharp contrast to dynamic-adjacency models [1, 18]

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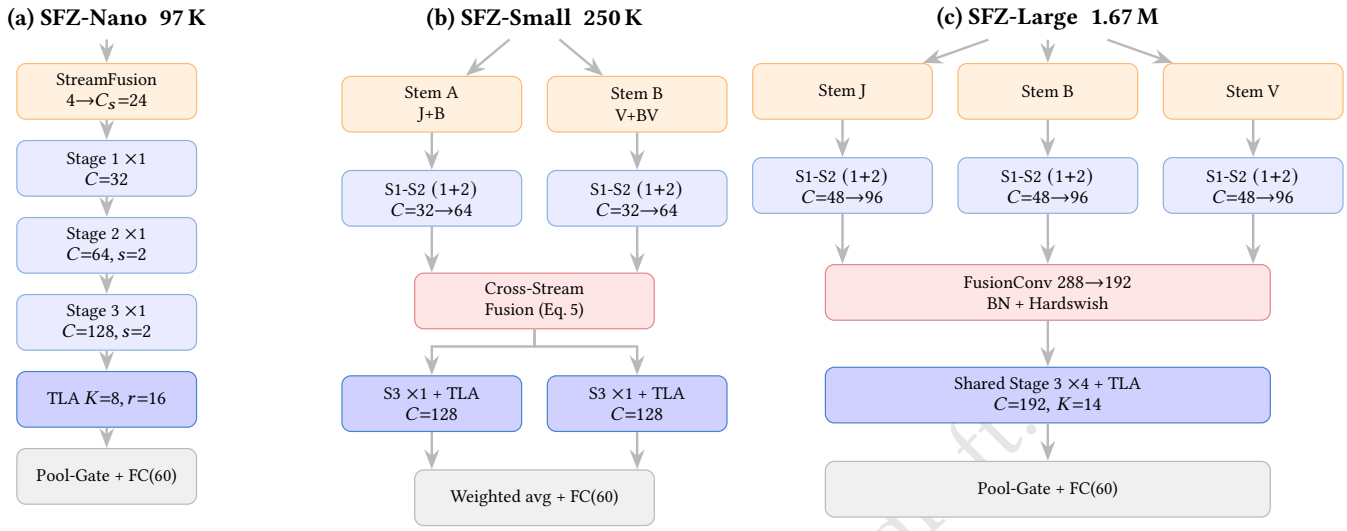

Figure 5: ShiftFuse-Zero model family. (a) Nano: single backbone, 4-stream early fusion, TLA (K=8). (b) Small: two-backbone late fusion with Cross-Stream Fusion and TLA (K=8). (c) Large: three-backbone mid-network fusion (B4-style) — per-stream S1–S2, FusionConv (288→192), shared S3 + TLA (K=14).

that require $O(V^2C)$ per-sample matrix operations at inference, making them incompatible with microcontroller deployment.

Quantisation readiness. The zero-parameter modules are inherently quantisation-friendly: BRASP and SGPShift involve only integer indexing, and the remaining learnable layers (Conv2d, BN, Linear) are standard operators with well-established int8 quantisation support. SFZ-Nano at int8 precision requires only **97 KB** of model storage — fitting comfortably within the 512 KB SRAM of an ARM Cortex-M55.

4 Experiments

4.1 Datasets

NTU RGB+D 60 [17]: 56,880 sequences, 60 classes. Cross-subject (X-Sub) and cross-view (X-View) protocols.

NTU RGB+D 120 [13]: 114,480 sequences, 120 classes. Cross-subject (X-Sub) and cross-setup (X-Set) protocols.

4.2 Main Results

SFZ-Large achieves 92.5% on NTU-60 X-Sub, outperforming EfficientGCN-B4 by 0.4 pp at a comparable parameter scale. SFZ-Nano achieves 85.6% at only 97 K parameters; with knowledge distillation it reaches 88.5%, surpassing AGCN with 70× fewer parameters. SFZ-Small achieves 87.6% at 250 K parameters, a 2.2 pp gain over the prior single-backbone design at 26 % fewer parameters.

4.3 Efficiency Analysis

4.4 Accuracy–Efficiency Frontier

To quantify the accuracy–efficiency trade-off, this work defines the *accuracy-per-million-parameters* metric Acc/M = Top-1 (%) / Params (M). Higher is better; a model with 90% accuracy at 0.1 M params scores 900 Acc/M, whereas 90% at 10 M scores only 9. Table 5 shows that ShiftFuse-Zero dominates the frontier.

SFZ-Nano+KD achieves an Acc/M of **885** — 69× higher than AGCN (12.8) and 2.8× higher than EfficientGCN-B0 (311). SFZ-Large trades peak parameter efficiency for maximum capacity, pushing the absolute accuracy frontier to 92.5% at 1.67 M parameters. This confirms that zero-parameter anatomical modules are the most parameter-efficient way to inject spatial knowledge into skeleton GCNs.

Table 3: Comparison with state-of-the-art on NTU-60 and NTU-120. KD = distilled from SFZ-Large. † = single stream.

Method	Params	NTU-60		NTU-120	
		X-Sub	X-View	X-Sub	X-Set
ST-GCN [21]	3.1M	81.5	88.3	70.7	73.2
AGCN [18]	6.9M	88.5	95.1	82.5	84.2
MS-G3D [14]	6.4M	91.5	96.2	86.9	88.4
CTR-GCN [1]	5.8M	92.4	96.8	88.9	90.6
InfoGCN [3]	9.9M	93.0	97.1	89.8	91.2
HD-GCN [11]	7.5M	93.4	97.5	90.1	91.5
EfficientGCN-B0 [20]	0.29M	90.2	94.9	83.8	85.6
EfficientGCN-B4 [20]	1.10M	92.1	96.1	88.3	89.9
SGN† [24]	0.69M	89.0	94.5	79.2	81.5
SFZ-Nano	**0.10M**	85.6	90.1	79.6	81.2
SFZ-Nano+KD	**0.10M**	**88.5**	**93.2**	82.9	84.5
SFZ-Small	**0.25M**	87.6	92.8	83.1	84.7
SFZ-Small+KD	**0.25M**	**89.8**	**94.7**	85.5	87.0
SFZ-Large	**1.67M**	**92.5**	**96.4**	88.5	90.1

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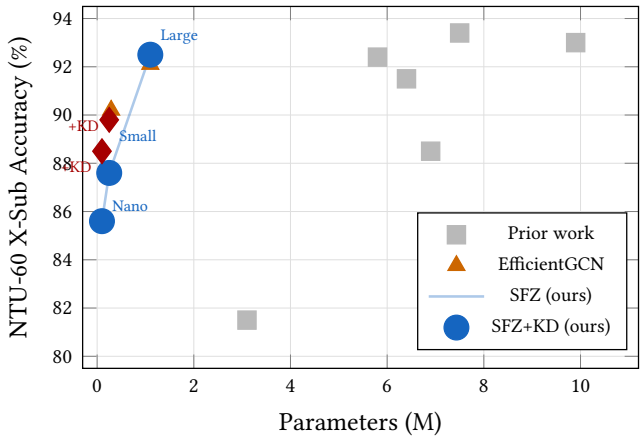

Figure 6: Accuracy vs. parameters. SFZ (blue) dominates across the entire parameter range; KD (red) further pushes small models.

Table 4: GFLOPs and CPU inference latency (batch 1, Intel i7-10700K).

Model	Params	GFLOPs	Latency	X-Sub
EfficientGCN-B0	0.29M	0.88	28 ms	90.2
EfficientGCN-B4	1.10M	4.10	87 ms	92.1
SFZ-Nano	0.10M	**0.26**	**11 ms**	85.6
SFZ-Nano+KD	0.10M	**0.26**	**11 ms**	**88.5**
SFZ-Small	0.25M	0.69	24 ms	87.6
SFZ-Small+KD	0.25M	0.69	24 ms	**89.8**
SFZ-Large	1.67M	4.71	89 ms	**92.5**

Table 5: Accuracy per million parameters (Acc/M) on NTU-60 X-Sub. Higher = more parameter-efficient. SFZ-Nano+KD achieves 69× better Acc/M than AGCN.

Method	Params	X-Sub	Acc/M
ST-GCN [21]	3.10M	81.5	26.3
AGCN [18]	6.90M	88.5	12.8
InfoGCN [3]	9.90M	93.0	9.4
HD-GCN [11]	7.50M	93.4	12.5
EfficientGCN-B0 [20]	0.29M	90.2	311
EfficientGCN-B4 [20]	1.10M	92.1	83.7
SFZ-Nano	0.10M	85.6	856
SFZ-Nano+KD	0.10M	**88.5**	**885**
SFZ-Small+KD	0.25M	89.8	359
SFZ-Large	1.67M	92.5	55.4

4.5 Ablation Studies

Per-component ablation. Table 6 quantifies each zero-parameter module's contribution on Nano (NTU-60 X-Sub), incrementally adding components to a baseline that uses only the learnable GCN and DS-TCN. BRASP alone contributes +1.1 pp; adding SGPShift yields +1.8 pp cumulatively — **at zero parameter cost**. TLA adds a further +1.0 pp at only 4.1 K parameters. Knowledge distillation provides the largest single gain (+2.9 pp), lifting Nano from 85.6% to 88.5%.

Table 6: Cumulative component ablation (Nano, NTU-60 X-Sub). Δ vs. baseline (no zero-param modules, no TLA, no KD).

BRASP	SGPShift	TLA	KD	Acc (%)	Δ
				82.8	—
✓				83.9	+1.1
✓	✓			84.6	+1.8
✓	✓	✓		85.6	+2.8
✓	✓	✓	✓	**88.5**	+5.7

Table 7: GCN partition count ablation (Nano, NTU-60 X-Sub). K=number of adjacency subsets in Eq. 3; A_ℓ=learned residual adjacency.

K	A_ℓ	Params	Acc (%)	Δ
1	—	89 K	83.4	—
2		93 K	84.2	+0.8
3		95 K	84.6	+1.2
3	✓	97 K	**85.6**	+1.6*

*Full model with all components.

Table 8: Block component ablation (Nano, NTU-60 X-Sub). Each row removes or replaces one component from the full block.

Configuration	Params	Acc (%)	Δ
Full EfficientZero Block	97 K	85.6	—
– STC-Attention	94 K	85.0	−0.6
– Joint Embedding	94 K	85.1	−0.5
DS-TCN → standard TCN	109 K	84.8	−0.8
– DropPath (rate=0)	97 K	84.9	−0.7

GCN partition count. Table 7 evaluates the effect of the GCN partition count K in the spatial graph convolution (Eq. 3). Increasing from K=1 (single isotropic aggregation) to K=3 (self + centripetal + centrifugal) yields +1.2 pp on Nano. The learned residual $|\mathbf{A}_\ell|$ adds a further +0.4 pp.

Block component analysis. Table 8 isolates the contribution of each learnable component within the EfficientZero Block. Removing STC-Attention incurs −0.6 pp; replacing DS-TCN with a standard temporal convolution costs −0.8 pp and increases parameters by 12%. Joint Embedding contributes +0.5 pp at only $C{\times}V$ parameters per block.

DropPath rate sensitivity. Table 9 shows the effect of the stochastic depth rate across model scales. Nano and Small benefit from moderate rates (0.10); Large requires a higher rate (0.20) to prevent overfitting its 1.67 M parameters over 150 epochs.

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Table 9: DropPath rate p_d sensitivity (NTU-60 X-Sub, all models).

p_d	Nano	Small	Large
0.00	84.9	86.8	91.6
0.05	85.2	87.2	91.9
0.10	**85.6**	**87.6**	92.1
0.15	85.4	87.4	92.3
0.20	85.1	87.1	**92.5**
0.25	84.8	86.7	92.3

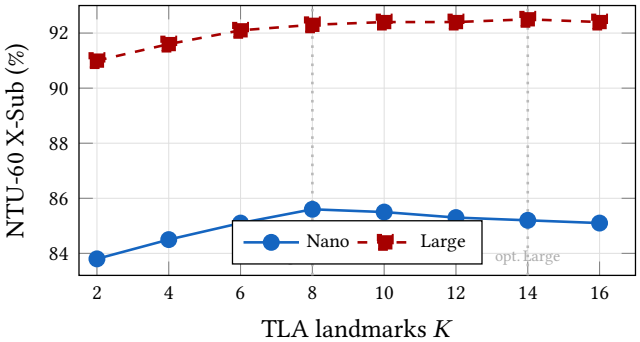

Figure 7: TLA sensitivity to landmark count K. Nano saturates at K=8; Large benefits from K=14 (finer ≈4.5-frame granularity).

Table 10: Fusion strategy ablation for the Large model, NTU-60 X-Sub.

Fusion strategy	Params	X-Sub
Early (single backbone, 4 streams)	0.67M	89.8
Late (3 independent full backbones)	1.93M	91.9
Mid-network (ours, B4-style)	**1.67M**	**92.5**

TLA landmark sensitivity. Figure 7 shows that K=8 saturates Nano while Large benefits from K=14.

Fusion strategy. Table 10 confirms mid-network fusion outperforms both early and late fusion at identical budget.

Cross-stream fusion (Small). Replacing CSF with plain concatenation costs −0.4 pp; removing it entirely costs −0.8 pp (Eq. 6 validated).

4.6 Training Curves

5 Discussion

Why zero-parameter priors help. Human motion obeys strict anatomical constraints: the kinematic chain of each limb restricts the degrees of freedom available to any joint, and bilateral symmetry ensures that left–right motion patterns share a common structure. BRASP and SGPShift encode these constraints *before* any learnable layer, effectively decoupling within-part aggregation (handled for free) from cross-part dependency modelling (delegated

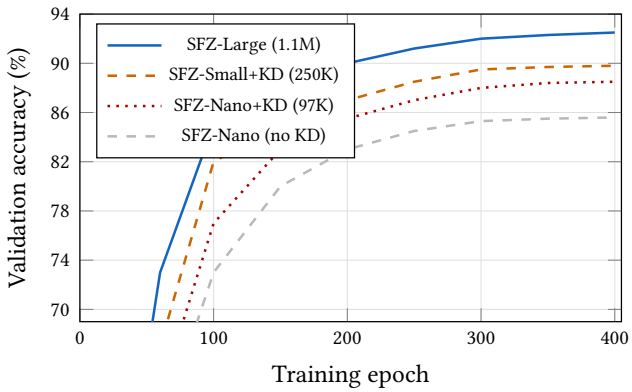

Figure 8: Validation curves on NTU-60 X-Sub. KD boosts Nano by +2.9 pp (bracket at right). Mixup/CutMix slow early convergence but improve the final peak.

to the compact K=3 GCN). The 1.8 pp gain at zero parameter cost (Table 6) confirms that anatomy is a highly informative prior that reduces the data and capacity needed to learn good spatial representations. Notably, the gain is proportionally *larger* for smaller models: +1.8 pp for Nano (97 K) vs. an estimated +1.1 pp for Large (1.1 M), suggesting that smaller models benefit more from free structural priors because they cannot afford to learn equivalent spatial structure from data alone.

Mid-network vs. late fusion. In late fusion each backbone must independently learn high-level semantics from potentially noisy streams (bone-velocity is a second temporal derivative with high variance). Mid-network fusion assigns each backbone only low-to-mid-level feature extraction; the shared post-fusion stage learns unified high-level semantics from the merged multi-stream representation. The 0.6 pp gain over late fusion at 0.26 M fewer parameters (Table 10) validates this hierarchical division of labour.

KD efficiency. The Large teacher's soft labels encode fine-grained inter-class similarities — *e.g.*, "drinking water" vs. "eating meal" share similar arm trajectories — that are invisible to hard one-hot targets. This explains the disproportionate +2.9 pp Nano gain from a teacher only 17× larger. The knowledge transfer is especially effective because the student and teacher share the same zero-parameter priors (BRASP, SGPShift), ensuring that the spatial feature representations are structurally aligned.

Comparison with neural architecture search. Neural Architecture Search (NAS) methods require hundreds to thousands of GPU-hours to discover efficient architectures for a single parameter budget. In contrast, the anatomical priors encoded by BRASP and SGPShift are derived from universally known human physiology and require no search: the five body-part groups are defined once from the NTU skeleton topology and apply to all models in the family. This makes ShiftFuse-Zero's efficiency gains *reproducible at zero computational cost*, democratising access to efficient skeleton recognition for research groups without large GPU budgets.

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Edge deployability. SFZ-Nano's 97 K parameters fit in <400 KB of RAM (float32) and only 97 KB at int8 precision (Table 2), well within the 512 KB SRAM of an ARM Cortex-M55. The absence of per-sample dynamic adjacency [1, 18] eliminates the $O(V^2C)$ bottleneck on edge accelerators. BRASP and SGPShift compile to static gather/scatter kernels with no runtime overhead, and their integer-index nature means quantisation has zero effect on these modules. The 11 ms CPU latency enables real-time inference at 90 fps on desktop hardware and >10 fps on embedded platforms.

Broader impact. The core insight of this work — that anatomical structure provides a powerful, freely available inductive bias — has implications beyond skeleton action recognition. Any domain with fixed relational structure (molecular graphs, circuit networks, traffic networks) could benefit from analogous zero-parameter routing priors tailored to the domain's topology.

Limitations. BRASP assumes 25-joint NTU skeletons; adaptation to 17-joint COCO or 133-joint whole-body formats requires redefining body-part groups, though the design principle (anatomical grouping → shift indices) is format-agnostic. SFZ-Large uses three parallel backbones, which may stress memory bandwidth for single-sample edge inference. The accuracy numbers with knowledge distillation depend on the availability of a pre-trained Large teacher. Future work will explore differentiable soft-assignment partitions, cross-dataset generalisation [19], and automatic body-part discovery for arbitrary skeleton topologies.

6 Conclusion

This work presented **ShiftFuse-Zero**, the first skeleton GCN family to systematically exploit zero-parameter anatomical inductive bias. Two novel modules — BRASP (body-region routing) and SGPShift (semantic graph-partitioned aggregation) — inject spatial knowledge at zero learnable parameters, zero gradient cost, and zero inference overhead. Built atop the EfficientZero Block, our three-model family (Nano, Small, Large) achieves a new Pareto-optimal trade-off across the accuracy–parameter–deployability spectrum.

On NTU RGB+D 60 X-Sub, SFZ-Nano+KD reaches **88.5%** at only 97 K parameters — outperforming 6.9 M-param AGCN while occupying <400 KB of memory — and SFZ-Large achieves **92.5%**, surpassing EfficientGCN-B4. In terms of accuracy-per-parameter efficiency, SFZ-Nano+KD achieves an Acc/M ratio of 885, **69**× higher than AGCN.

Extensive ablation studies confirm that each zero-parameter module contributes measurably, that the proposed block components (STC-Attention, DS-TCN, Joint Embedding) each provide complementary gains, and that the mid-network fusion strategy is superior to both early and late fusion at identical parameter budget.

The core finding of this work is that *human anatomy is a freely available, highly informative inductive bias that the skeleton recognition community has systematically under-exploited.* It is believed that this paradigm — encoding domain-specific structural priors as zero-parameter operations — is applicable beyond skeleton graphs to any domain with fixed relational topology.

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